

Bomb Disposal

?????? ?? ?????????????????????????? ? ?????????? ??? ????????

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???? ???? ?????? ????????????, ? ?????????????? ?????? ??????: ????????? ??? ?? 2 ??????, ????????? ????
 ?????, ??? ?? ?????? ? ?????????, FishNet RPC/SyncVar, UI, ??????, Linux dedicated server ? ??????????
 ??????????

1. ??? ?? ?????

?????: ?????????????????? ???? ?? ?????. ?????? ?????????????? ? ??? ???? , ????????,
 ????????????? ???? ?? ? ???? ? ?????????? ???? ? ???? ???? ? ???? ?????. ??? ???? 2 ??????
 ????????? ???? ?? ???? ?? ???? ????.

- `?????: ????????` PvP/objective-based multiplayer mode.
- `???????????:` Unity, FishNet, SyncVar, ServerRpc, ObserversRpc, broadcast `??????????`, CSP/reconcile.
- `???????????:` Host/Client `???` `???????????` `??????`; `?????????????` `???????????` Dedicated Server `???` Linux/headless.
- `???????????` `?????:` `???????????` `???????????` `?????????????????`, `?????????????` `?????????????` `?` `?????????????????` `???????????` `?????`.

2. ??????? ???? ? ????????

???	?? ?????	?? ???? ?????
???	????? ???? ???? ???? ?????, UI ????????? ? ??????? ? countdown.	?? ??????? lifecycle: ?????? ? ??? ? ?????????
???? ?????	????? ????????? ?????, ??????? ???? ? ??????, ????????, ???? ? ? ???? ? Cover_.*.	????? ??????? ? ?????, ? ? ?????.
?????? ?????	???? ????????? ???? , ????????? 10-???????? fuse, ???????? ???? ????.	??? ??? ? ????????? ???? , ??????? ? ???????? ?????
???/????	??? ? ??????? ??????? ? fuse ???? , ???? ????????? ? ?????????	????? ??????? ????????? ? ????????? ????????.
?????	??? ??????? ???? ? ???? ?????????, ??????? ???? , ???? ???? ? ????.	? ? ??????? ???? ???-????.
?????????	???? 2 ???? ?????????? ?????????/? ? ???? ? ????.	????? ????????? ????????? ????.

3. ??????????? ?? : ??? ??????????????

???? ??	?? ????????	?? ???? ? ??	????
???? ? ???????????	Host/Client ??? Dedicated, NetworkManager, UI ??????????.	FishNet NetworkManager, ?????? Host/Client, ????? Server/Client ??? dedicated.	20
State Sync	HP, ??, ???, ??????? ?? UI.	SyncVar: Nickname, HP, IsAlive, Score; GameStateBroadcast ??? ?????/????.	15
????????/ownership	???????? ?????? ?????????? ?????? ?????, ?????? ??	PlayerMovement ? Replicate/Reconcile; PlayerCamera ?????? ??? owner.	15

	?????????		
RPC	????????? ????????????????	ServerRpc ??? ??????, ??????/?????/?????; ObserversRpc ??? ????????/?????	15
?????? ????	???? ? ??? ? ??????????	GameState: WaitingForPlayers, InProgress, ShowingResults.	10
?????????	????? ????????? ? ? ??????	GameManager, PlayerNetwork, PlayerMovement, PlayerShooting, ConnectionUI, SceneBootstrap.	10
UI/?????	????????? ????, HUD, HP, status.	??????? ????, ???/?????, labels, ping/FPS/CSP overlay, ??????	5+
????????????/???????? ?????????	30 FPS, ??? ?????????? ??????	FPS overlay, ?????? ?????? ? ? ?????? ??? ???? ???? ???? ?????/??????	5
????????????	??? ???? ? ??????????	Bomb Disposal ? object scoring, random objective positions, death debris.	5

4. ??????????: ???? ??????? ?????

?????	??? ???????
GameManager.cs	?????? ?????????? ?????: ??? ?????, ????, ???, ?????, ????, random spawn, broadcast ?????????
PlayerNetwork.cs	SyncVar ?????: Nickname, HP, IsAlive, Score; ?????, ????? ?????, ????????? ????? ? ?????
PlayerMovement.cs	FishNet Replicate/Reconcile, ?????????? ?????????????, ????????? CharacterController, ?????? ? ???????
PlayerShooting.cs / Projectile.cs	???????? ????? ServerRpc, ?????? SyncVar, ?????, ????????? ? ? ?????? ??????
ConnectionUI.cs	??? ??????????, HUD, ping, FPS, CSP toggle, ????????? ? ? ??????
FinalProjectSceneBootstrap.cs	???????/????????? ????? ? ??????, ???, ?????, ????????? ?????? ????? ? ??????????
ServerAutoStart.cs	????????? ?????? ??? dedicated/headless ??????

5. ??????, ?????? ?????????? ???? ???????

??? ? ????????????

?????	???????? ?????	??? ? ??????
????? FishNet, ? ? NGO?	FishNet ??? ????????????? ? ??????? 3/4, ??? SyncVar, ServerRpc/ObserversRpc, prediction/reconcile ? ?????? dedicated server workflow.	??? ??????
???? ? ??? ??????????	??? ????? Host/Client: ??? ?????? ????????????? ????? ? ?????. ??? ????? ?????? ????????????? Linux Dedicated Server: ????????? headless server, ?????? ????????????? ? ????	NetworkManager , ServerAutoStart
??? ?????? ?????? ??????????????	????? ?????????? ??????????: ?????????, ?????????, ?????????? ????? ?????????? ??????, ?????????? ?????, ?????????? ????? ? ????? ?????? ?????? ??????????	GameManager
??? ?????????????????? ????? SyncVar?	? ??????: ???, HP, alive, score. ? ??????: ammo. ??? ?????? ????????? ?????? ?????????????? ?????????? broadcast, ?????? ??? ??? ?????????????????? ??????	PlayerNetwork, PlayerShooting
????? BombPosition ?? SyncVar?	???? ? ? ?????????? NetworkObject ? ?????, ? ?????????? ????????? objective-state. ?????? ?????? ?????????? GameStateBroadcast: ???, ??????, carrier, fuse, objective text.	GameManager

Gameplay ? ????????? ??????

??????	????????? ?????	??? ? ???????
??? ?????????? ?????	??? ?????? ?????? ?????, ?????? ?? ? ?????????? ?????????? carry score. ? ? ?????????? ?????? ?????? ?????????? ??????. ?????????? ??? ? ? ???.	GameManager
?????? ?????????? ? ???? ?????	????? ?????? ????: ?????? ?????? ?????? ?????? ?????, ? ? ?????????? ????? ? deathmatch.	????????? ??????
??? ?????????? ?? ?????????? ? ?	?????? ??????????, ?? ???? ?????????? ??????? BombCarrier; ?????? ?????????? ?????????? ????? ? ?????? ?????? ?? respawn.	PlayerShooting, GameManager
??? ?????????? ????????	????????? ?????? ???, ??? ? ?????? ???, ????? ?????? ??? ? ?????? disposal zone. ?????? ?????????? ?????? ?????????? ????? ? ?????????? ?????.	GameManager
?????? ?????? ?? ?????????? ? ???????	????? ??????? ?????? ?????? ?????????? ?????????? Cover_* ? clearance ??? ??????/????????; ??? ?????? ??????, ?????????? ??????	GameManager
?????? ??????? ?????? ? ?????????? ?	PlayerNetwork ?????????? PlayerSpawn ? ?????????? ? ??????? Cover_*; ??? ???? ??????, ?????? fallback-????????.	PlayerNetwork

????????, prediction ? ????????

??????	????????? ?????	??? ? ???????
??? ????? CSP?	Client-side prediction: ?????????? ?????? ?????? ?????????? ????, ?? ??? ? ?????? ?????????? ?????? ?????? ?????????? reconcile, ?????? ?????????? ?????????????.	PlayerMovement
??? ?????? Replicate/Reconcile?	Replicate ?????????? ???/? ?????????? ?????????, Reconcile ?????????? ?????????????? ????????? ?????????.	PlayerMovement
?????? ?????? ?????? ?????????? ? ?	?????? ?????? ??? BombPosition ? ? ?????????????? broadcast. ??????????: ? Carried-????? ?????? ?????????? ? ??????? ????????? PlayerNetwork ?????????.	FinalProjectSceneBootstrap
?????? ? ???? ???? ?????? ??????	??? ?????????? ?????????????? ??????: ?????? ?????????? ?????? ?????????????? ?????????????, ? ?????? ?????????? ??????? ?????????? ? ? ??????	GameManager, SceneBootstrap

UI, ?????, ??????????????

??????	????????? ?????	??? ? ???????
??? ?????????? UI?	Mode, ping, ???/CSP, ?????????? ?????, ?????? ?????????, ?????, nickname, HP, score, ammo, ?????? ????? ? ?????????? ?????????.	ConnectionUI
??? ?????? ??????????????????????	? ?????? ??????? ???? ???? FPS overlay; ping ?????????? ? HUD.	ConnectionUI, SceneBootstrap
??? ??????? ?? ??????????	????????? ?????, ?????????? ??????, ???????, ?????? ??????, ?????? ??????/????????, ??????? ??????, ?????, death debris.	SceneBootstrap, PlayerNetwork
??? ?????????? ?????????? ???????	??? IsAlive=false ??????? MeshRenderer-????? ????????????? ? ??????? ? Rigidbody/BoxCollider ? ?????????????? ??? ????????? debris ??????????	PlayerNetwork
??? ?????? ?? ?????????? ??????????	?????? ?????? ?????? carrier/????? ? ?????????? ????? ? ??????/respawn, ?????? ??? ? ??????	GameManager

??????	???????? ?????	??? ? ??????
????? WSL/Linux server?	????? ?????????? dedicated/headless ?????? ????? ? ?????????? ????????? ??????????: ?????? ??????????, ??????? ??????????.	ServerAutoStart, build
??? Host ?????????? ?? Dedicated?	Host ??? ?????? ? ?????? ??????. Dedicated ?? ??????, ?????? ????????????? ??? ? ?????????????? ??????.	Network topology
??? ?????????? ?? ??????	??? ???????/??: ?????????????, ????? ?????, ?????? ?????, ????????, ??????????/??:, ??????, ??????????? score, ???????????.	demo plan
??? ??????, ??? ? ?????? ?? ??????	???????, ?? ??? ? ??????????: ????? ? ???, ????? ? ??????, ?? ?????? ? ???, ??????????? objective ? cover; ?? ?????????? ????????????? ? ?????????.	git history

- ?????? GameManager ??????? ?? ?????????? ?????? ?????, ?? ?????????? ?????? ??????????: ??????????, ?????????, UI, ??????? ? ?????????? ?????? ?????????? ?????????.
- ?????? ????? ?? NetworkObject? ?????? ??? ? ??? ?????? ?????????????????? ?? ?????????? ? ???????, ? ?? ?????????? ?????????????? ?????????? ???????? ??????
- ?????? ??? ?? ?????????? ?????? ?????? ?? ??? ?? ??? ?????????? score.
- ?????? random spawn ?? ?????????? ?????? ? ???? ?????????? ???? ?????????????? ??????? ? ?? ???? ??????????????
- ?????? ???? broadcast ??????? SyncVar? SyncVar ??????? ??? ?????????? ?????????????? NetworkBehaviour, ? broadcast ??????? ?? ??????? HUD-????????? ??????
- ??? ?????????????? ?????? ?? ?????????????? ??????: score ?? ???????, fuse, ??????-?????, ?????? ?? ??????????, random objective ? ?????????? death debris.

1. ??????? ?????????? ????: ??? Bomb Disposal, ????????? multiplayer-????? ?????????.
2. ?????????? ?????????????? ??? ? ????????? ? HUD: ping, state, players, time, HP, score.
3. ?????????? ????: waiting ? match ? bomb pickup ? fuse ? throw/hit/explosion ??? defuse ? results.
4. ?????????, ??? ??????? ??????????????: ??? ??????? ?????????? ?????????? ?????? ServerRpc ? ?????????? GameManager.
5. ??????????, ?????? ??????? SyncVar, ? ?????? ??? ? ?????? GameStateBroadcast.
6. ?????????? ?????????? ??????: CSP, Linux dedicated server, FishNet, ??????/?????????, FPS overlay.

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? ??????? ? ? ? ??????????????: PlayerNetwork ?????? ??????? ????????? ??????, PlayerMovement ??????? ?
?????????????? ?????????, PlayerShooting ? ? ?????? RPC, ConnectionUI ? ? ?????????, ? GameManager ??????
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